

Selective HER Activation of Monolayer CrCl_3 by Transition-Metal Functionalization

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Supporting Information Available

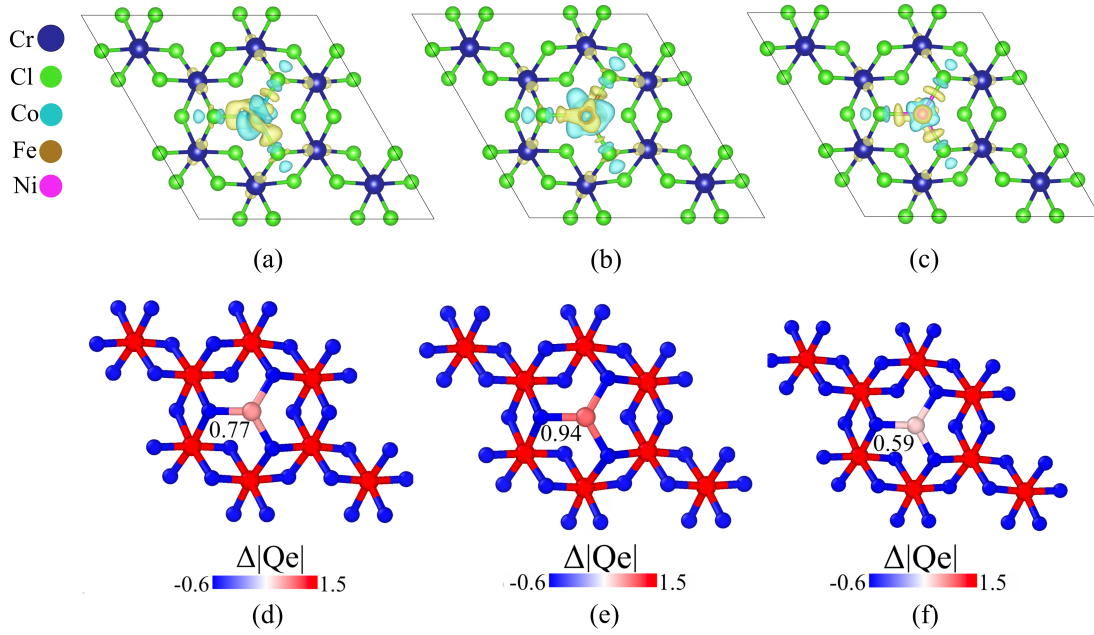


Fig. S1. Top views of the charge-density-difference isosurfaces for (a) Co-, (b) Fe-, and (c) Ni-adsorbed CrCl₃, respectively, plotted at an isosurface value of $\delta = 5^{-3}$ eV/Å³. The two colors denote regions of charge accumulation and charge depletion induced by adatom adsorption. (d-f) Top views of the corresponding Bader charge analysis for Co-, Fe-, and Ni-adsorbed CrCl₃, respectively. The values indicated in each panel represent the magnitude of the charge transfer associated with the adsorbed transition-metal atom, in units of $|e|$.

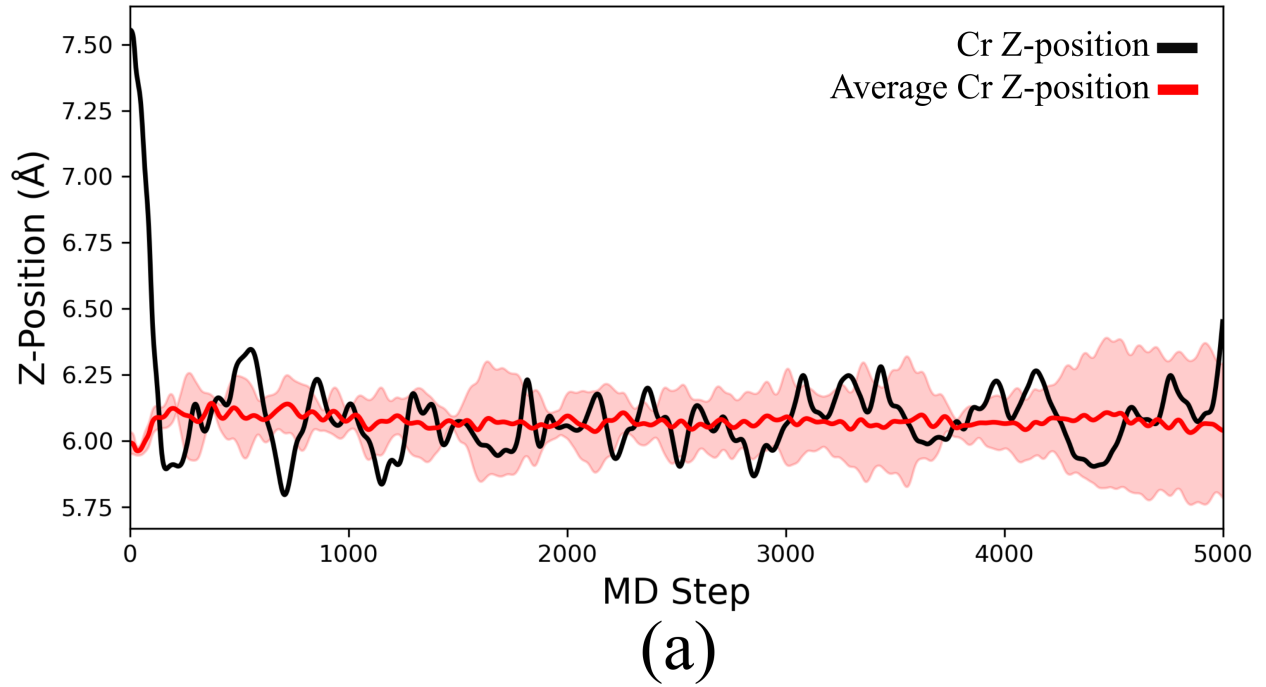


Fig. S2. Time evolution of the total energy (blue curves, left axis) and instantaneous temperature (red curves, right axis) obtained from the AIMD simulations performed at 300 K for the CrCl_3 monolayer with Cr adsorbed at the hollow site.

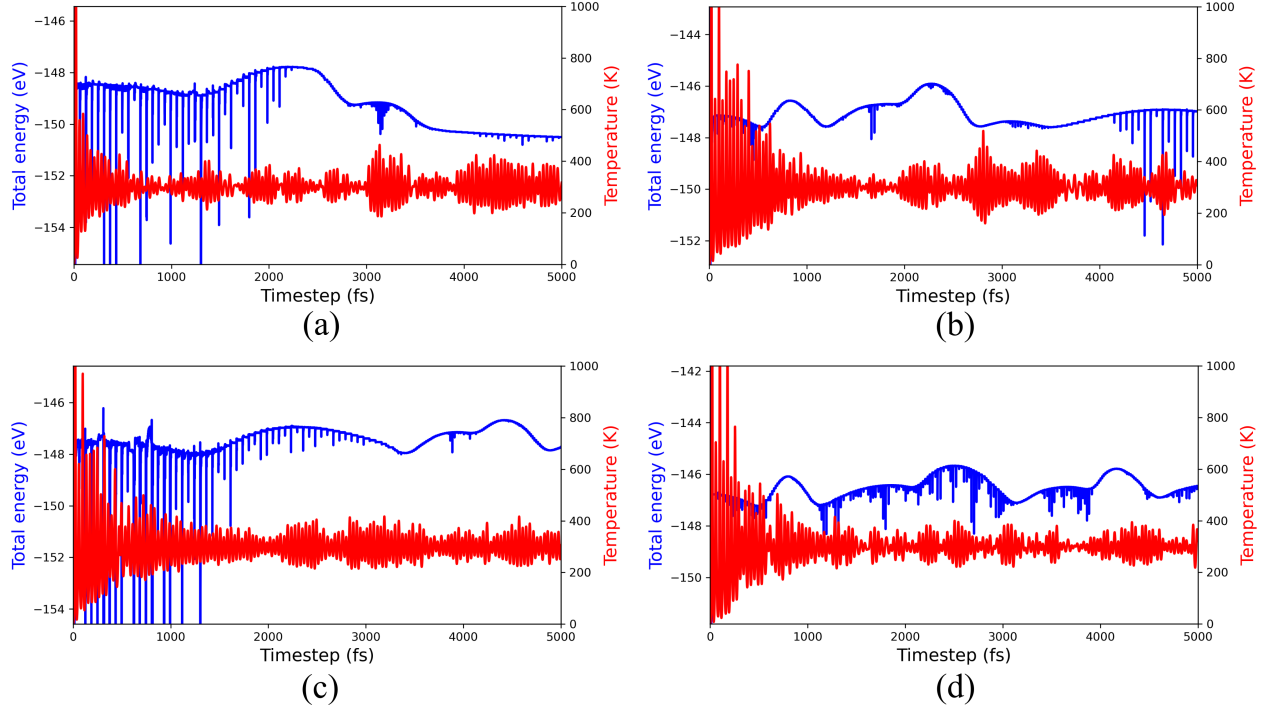


Fig. S3. Time evolution of the total energy (blue curves, left axis) and instantaneous temperature (red curves, right axis) obtained from the AIMD simulations performed at 300 K for the CrCl_3 monolayer with adsorbed (a) Cr, (b) Co, (c) Fe, and (d) Ni atoms at the hollow site. The results confirm the thermal equilibration of all systems during the 5 ps simulation window.